

Yunhan Qiao

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RESEARCH INTERESTS

My research sits at the intersection of empirical software engineering, code comprehension, and human-AI collaboration. I study how students and experienced developers work with GenAI coding assistants and agents (e.g., GitHub Copilot, Cursor) in realistic *brownfield* codebases, and whether faster, more correct work also means deeper understanding. I combine controlled experiments, screen-recorded behavioral analysis, prompt coding, EEG, and interviews to inform how computing education teaches and assesses GenAI-assisted development and how future AI coding agents and IDEs should be designed.

EDUCATION

- **Ph.D. in Computer Science, School of EECS** Oregon State University, Corvallis, OR
Ph.D. Candidate (passed preliminary exam, Jun 2026) Sep 2023 – Jun 2027 (expected)
 - **Advisor:** Dr. Christopher Hundhausen at the [Software Engineering Education Assessment and Analytics \(SEAL\) Lab](#)
 - **Collaborator:** Dr. Anita Sarma (EEG study of GenAI-assisted programming)
 - **Research Areas:** Empirical Software Engineering, Code Comprehension, Computing Education, Human-AI Interaction
- **M.S. in Software Engineering** Arizona State University, Tempe, AZ
Thesis track Aug 2021 – May 2023
 - **Advisor:** Dr. Robert LiKamWa
 - **Thesis:** Networked System for Volumetric Athletic Coaching in Augmented Reality
- **B.E. in Software Engineering & B.E. in Information Technology (Dual Degree)** Liuzhou, China
Guangxi University of Science and Technology & Southern Cross University Sep 2016 – Jun 2020

PUBLICATIONS

Journal Articles & Conference Papers

- Yunhan Qiao, Md Istiak Hossain Shihab, and Christopher Hundhausen. **A Systematic Literature Review of the Use of GenAI Assistants for Code Comprehension: Implications for Computing Education Research and Practice.** *TOCE: ACM Transactions on Computing Education*, 26(2), Article 24, 1–33, 2026. [\[PDF\]](#) [\[DOI\]](#)
- Ahsun Tariq, Phillip Conrad, Christopher Hundhausen, S. Chandrasekaran, Yunhan Qiao, and Summit Haque. **Using Slackbot-Prompted Reflections to Promote Participation and Reduce Contribution Inequalities in Team Software Development Projects.** *TOCE: ACM Transactions on Computing Education*, 2026.
- Md Istiak Hossain Shihab, Christopher Hundhausen, Ahsun Tariq, Summit Haque, Yunhan Qiao, and Brian Wise Mulanda. **The Effects of GitHub Copilot on Computing Students' Programming Effectiveness, Efficiency, and Processes in Brownfield Coding Tasks.** *ICER'25: Proceedings of the 2025 ACM Conference on International Computing Education Research*, Vol. 1, 407–420, Aug. 2025. [\[PDF\]](#) [\[DOI\]](#)
- Manish Motwani, Aakash Kulkarni, Yunhan Qiao, Matthew Davis, and Ziyang Chen. **LLM-Guided Differential Fuzzing for Detecting Platform-Specific Bugs in Scientific Applications.** *AIxSE'25: IEEE International Conference on AI x Software Engineering*, 2025. [\[PDF\]](#)

Under Review

- Yunhan Qiao, Md Istiak Hossain Shihab, Summit Haque, and Christopher Hundhausen. **Faster but Not Wiser: GitHub Copilot Decouples Programming Performance from Code Comprehension in Brownfield Tasks.** *TOCE: ACM Transactions on Computing Education*, under review. [\[arXiv\]](#)
- Yunhan Qiao, Summit Haque, and Christopher Hundhausen. **Beyond the Prompt: Linking What Developers Ask, Do, and Understand with Coding Agents.** *SANER'27: IEEE International Conference on Software Analysis, Evolution and Reengineering*, under review. [\[arXiv\]](#)

Under Submission

- O. Schutz, Yunhan Qiao, F. Moussaoui, R. Choudhuri, and A. Sarma. **Heart and Mind in the Loop: Why We Vibe When We Work with AI.** *CHASE'27: International Conference on Cooperative and Human Aspects of Software Engineering*, under submission.

RESEARCH EXPERIENCE

- **Brain Activity of Vibe Coding on Developers: An EEG Study** Feb 2026 – current
Controlled EEG study of GenAI assistants (Cursor, GitHub Copilot) in brownfield tasks supervisor: Dr. Anita Sarma
 - Co-designed a controlled EEG study on an 88K-LOC MERN-stack codebase; ran in-person sessions with 30 developers, collecting neurophysiological data during vibe coding.
 - Applied Information Foraging Theory to model shifts in information seeking under the Ask and Agent modes of GenAI, and related code comprehension, cognitive effort, and task performance.
- **How Expert Developers Use GitHub Copilot in Brownfield Tasks** Jul 2025 – current
SEAL Lab; Beyond the Prompt (SANER'27, under review) supervisor: Dr. Christopher Hundhausen

- Designed a mixed-methods study of 10 expert developers on an 88K-LOC MERN-stack codebase; collected and analyzed 30+ hours of sessions via AWS WorkSpaces.
- Built the *Say, Do, Understand* workflow and a behavioral framework combining Information Foraging Theory and state machines; identified patterns of AI reliance, verification, and information seeking that shape task quality and codebase understanding.
- **GitHub Copilot's Effects on Students in Brownfield Programming Tasks** *Mar 2024 – Jun 2025*
SEAL Lab; ICER'25; Faster but Not Wiser (TOCE, under review) supervisor: Dr. Christopher Hundhausen
 - Designed a controlled experiment with GitHub Classroom on a 3K-LOC JavaScript codebase and a multidimensional comprehension assessment (system objective, implementation, debugging, reverse engineering, open-ended reasoning).
 - With 15 students, found a 50.7% reduction in completion time and a 71.4% increase in correctness with Copilot, but a comprehension–performance trade-off; active verification of AI-generated code correlated with stronger comprehension.
- **Systematic Literature Review of GenAI for Code Comprehension** *Sep 2024 – Jun 2025*
SEAL Lab; TOCE 2026 supervisor: Dr. Christopher Hundhausen
 - Analyzed 31 papers via thematic analysis and synthesized five GenAI approaches (e.g., software explanation, readability enhancement), benchmarking their effectiveness and evaluation methods.
- **Reflection and Equitable Participation in Team Software Projects** *Feb 2025 – Oct 2025*
SEAL Lab; TOCE 2026 supervisor: Dr. Christopher Hundhausen
 - Conducted qualitative analysis evaluating a Slackbot intervention designed to improve contribution equity in software teams.
- **LLM-Guided Differential Fuzzing for Scientific Software** *Jul 2023 – Sep 2024*
AIxSE'25 supervisor: Dr. Manish Motwani
 - Developed a Python LLM-guided seed-input generator; ran experiments on an HPC cluster and detected 8 of 20 heterogeneous bugs in LAMMPS.

TEACHING EXPERIENCE

- **Graduate Teaching Assistant, School of EECS** Oregon State University
Undergraduate software engineering courses *2024 – current*
 - **CS 362:** Software Engineering II
 - **CS 461:** Senior Software Engineering Project
 - **CS 467:** Online Capstone Project
 - **CS 492:** Mobile Software Development

PROFESSIONAL EXPERIENCE

- **Software Engineering Intern, Mocha Software** Nanjing, China
Software development *Aug 2020 – Nov 2020*

PROFESSIONAL SERVICE

- **Program Committee:** SIGCSE TS 2027, the 58th ACM Technical Symposium on Computer Science Education

SKILLS & COURSEWORK

- **Research Methods:** Controlled experiments, mixed methods, EEG studies, behavioral & log analysis, prompt coding, interviews, thematic analysis, survey design (Qualtrics), statistical analysis
- **Programming:** Python, Java, C, JavaScript, HTML, CSS, SQL, Bash
- **Frameworks & Tools:** React, Express, Flask, MongoDB, PostgreSQL, Git, AWS, Docker, $\text{\LaTeX}$; Claude Code, GitHub Copilot, Cursor, OpenAI Codex
- **Coursework:** Machine Learning; NLP with Deep Learning; Software Engineering for ML; Program Analysis; Algorithms & Data Structures; Human–Computer Interaction; Field Studies in SE & HCI; Methods of Data Analysis
- **Languages:** English (fluent), Mandarin Chinese (native)